

**K. J. Somaiya College of Engineering, Mumbai-77**

(Autonomous College Affiliated to University of Mumbai)

Semester: July 2014 - November 2014

**Max. Marks: 100**

**Duration: 3 Hrs**

**End Semester Exam**

Class: SE

Name of the Course : Applied Mathematics III

Course Code: UITC301

Semester: III

Branch: Information Technology  
& Computer

**Instructions:**

(1) All Questions are Compulsory

(2) Figures to right indicate full marks. Each sub-question has equal marks.

| Question No. |                                                                                                                                                                                                            | Max. Marks |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>Q.1</b>   | <b>Attempt any THREE of the following</b>                                                                                                                                                                  | <b>21</b>  |
| (a)          | Using Laplace transform evaluate $\int_0^{\infty} \frac{e^{-t} - \cos t}{te^{4t}} dt$                                                                                                                      |            |
| (b)          | Find Laplace transform of $f(t) = \frac{e^{-2t} \sin 2t \cos ht}{t}$                                                                                                                                       |            |
| (c)          | Using Convolution Theorem find inverse Laplace transform of $\frac{s}{(s^2+4)(s^2+1)}$                                                                                                                     |            |
| (d)          | Express the function $f(t) = \begin{cases} \sin t, & 0 < t < \pi \\ \sin 2t, & \pi < t < 2\pi \\ \sin 3t, & t > 2\pi \end{cases}$ as Heaviside's Unit Step function and hence find it's Laplace Transform. |            |
| (e)          | Solve using Laplace Transform: $(D^2 + 9)y = 18t$ where $y(0) = 0$ ; $y\left(\frac{\pi}{2}\right) = 0$                                                                                                     |            |
| <b>Q.2</b>   | <b>Attempt any THREE of the following</b>                                                                                                                                                                  | <b>21</b>  |
| (a)          | Find Fourier Series of $f(x) = \left(\frac{\pi-x}{2}\right)^2$ in $(0, 2\pi)$ and Hence deduce that $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$             |            |
| (b)          | Find half range cosine series of $f(x) = \sin x$ in $0 < x \leq \pi$ .<br>Hence deduce that $\frac{\pi^2-8}{16} = \frac{1}{1^2 \cdot 3^2} + \frac{1}{3^2 \cdot 5^2} + \frac{1}{5^2 \cdot 7^2} + \dots$     |            |
| (c)          | Find fourier series of $f(x) = x - x^3$ in the interval $(-1, 1)$                                                                                                                                          |            |
| (d)          | Obtain complex form of Fourier Series for $f(x) = e^{ax}$ in $[-l, l]$                                                                                                                                     |            |
| (e)          | Prove that $f_1(x) = 1$ , $f_2(x) = x$ , $f_3(x) = \frac{3x^2-1}{2}$ are orthogonal over $(-1, 1)$                                                                                                         |            |

|     |                                                                                                                                                                                                                                                                                                                                  |    |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Q.3 | Attempt any THREE of the following                                                                                                                                                                                                                                                                                               | 21 |
| (a) | Prove that $u = x^2 - y^2$ , $v = \frac{-y}{x^2+y^2}$ both are harmonic functions but $u + iv$ is not an analytic function.                                                                                                                                                                                                      |    |
| (b) | Verify Laplace's equation for $u = \left(r + \frac{a^2}{r}\right) \cos\theta$ . Also find the corresponding analytic function and imaginary part $v$ .                                                                                                                                                                           |    |
| (c) | If the imaginary part of the analytic function $w = f(z)$ is $v = x^2 - y^2 + \frac{x}{x^2+y^2}$ , show that the real part $u = -2xy + \frac{y}{x^2+y^2} + c$ .                                                                                                                                                                  |    |
| (d) | Find bilinear transformation which maps $z = 2, i, -2$ onto $w = 1, i, -1$ . Also find its fixed points.                                                                                                                                                                                                                         |    |
| (e) | Show that the transformation $w = i \left(\frac{1-z}{1+z}\right)$ maps<br>(i) the circle $ z  = 1$ onto the real axis of the W- plane and<br>(ii) the interior of the circle $ z  < 1$ onto the upper half of the W- plane.                                                                                                      |    |
| Q.4 | Attempt any THREE of the following                                                                                                                                                                                                                                                                                               | 21 |
| (a) | If the vectors $\vec{u}, \vec{v}, \vec{w}$ are non-coplanar, show that the vectors $\vec{u} \times \vec{v}, \vec{v} \times \vec{w}, \vec{w} \times \vec{u}$ are also non-coplanar. Hence, obtain the scalars $l, m, n$ such that $\vec{u} = l(\vec{v} \times \vec{w}) + m(\vec{w} \times \vec{u}) + n(\vec{u} \times \vec{v})$ . |    |
| (b) | Find the values of $a, b, c$ if the directional derivative of $\phi = ax^2y + byz + cz^2x^3$ at $(1, 2, -1)$ has maximum magnitude 64 in the direction parallel to the $z$ axis                                                                                                                                                  |    |
| (c) | Find $f(r)$ , so that the vector $f(r)\vec{r}$ is both solenoidal and irrotational.                                                                                                                                                                                                                                              |    |
| (d) | Verify Green's Theorem in the plane for $\int_C (xy + y^2) dx + x^2 dy$ where $C$ is the closed curve of the region bounded by $y = x$ and $y = x^2$ .                                                                                                                                                                           |    |
| (e) | Use Gauss's Divergence Theorem to evaluate $\iint_S \vec{N} \cdot \vec{F} ds$ where $\vec{F} = 4x\vec{i} - 2y^2\vec{j} + z^2\vec{k}$ and $S$ is the region bounded by $x^2 + y^2 = 4$ , $z = 0, z = 3$ .                                                                                                                         |    |
| Q.5 | Attempt any TWO of the following                                                                                                                                                                                                                                                                                                 | 16 |
| (a) | Find the Z- transform of $\{k^2 e^{-ak}\}, k \geq 0$ by using properties of Z- transform.                                                                                                                                                                                                                                        |    |
| (b) | Find the Z- transform of $f(k) = \sin(ak), k \geq 0$ hence find the Z- transform of $g(k) = 3^k \sin\left(\frac{\pi}{6}k\right), k \geq 0$ by using properties of Z- transform.                                                                                                                                                  |    |
| (c) | Find the inverse Z- transform of $f(z) = \frac{z}{(z-3)(z-2)}$<br>If R.O.C. is (i) $ z  < 2$ (ii) $2 <  z  < 3$ (iii) $ z  > 3$                                                                                                                                                                                                  |    |